



Statistical Learning Project

Davide Gamba, Matr. 1053470
Aurora Zanenga, Matr. 1054891

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1 Introduction

The main purpose of this project is to apply the knowledge gained from the Statistical Learning course in a practical manner by analyzing an interesting dataset about diamonds. This dataset is chosen due to its rich features that offer insights into diamond pricing.

The primary objective is to explore and identify the most efficient models for predicting diamond prices. We aim to determine whether the relationship between features and price is linear and to identify the variables that most significantly influence the price.

2 Preliminary Analysis

2.1 Dataset Description

The Dataset we've chosen for this project is a dataset that contains informations about Diamonds.

This dataset contains information regarding the prices and other attributes of nearly 54.000 diamonds.

It stands as a significant resource for those delving into data analysis and visualization for the first time.

Among the included data are:

Price	expressed in US dollars, ranging from \$326 to \$18,823
Cut	indicating the quality of the diamond's cut, with values including Fair, Good, Very Good, Premium, and Ideal
Color	denoting the diamond's hue, on a scale from J (worst) to D (best)
Clarity	serving as a measure of the diamond's clarity, rated from I1 (worst) to IF (best)

Table 1: Variables

Physical dimensions: provided as follows in millimeters:

Length (x)	ranging from 0 to 10.74 mm
Width (y)	ranging from 0 to 58.9 mm
Depth (z)	ranging from 0 to 31.8 mm
Depth%	is the depth of the diamond divided by the diameter of the diamond (43%–79%)
Table	is the length of the table facet divided by the diameter of the diamond(43%–95%)

Table 2: Cathegorical Variables

These data provide a detailed insight into the characteristics of diamonds, enabling in-depth analysis in the fields of valuation and trade of these precious gems.

2.2 Preprocessing Steps

The preprocessing steps for the dataset in this project involve several operations to ensure the data is clean, consistent, and ready for analysis.

Some columns names are updated for better clarity: the fifth column is renamed to "depth_percentage", the eighth to "length", the ninth to "width", and the tenth to "depth".

In order to have more significant and interpretable results, it was decided to align the dataset based on the price, so the dataset was restricted by taking into consideration only diamonds with a price between \$2000 and \$8000. Prices were converted from dollars to thousands of dollars to simplify numerical handling.

Categorical variables, such as cut, color, and clarity, were transformed into factors, essential for statistical modeling and machine learning algorithms.

The dataset was inspected for any missing values ensuring that there are no missing data points; outliers were detected and removed to ensure the data quality.

So the dataset used in analysis contains 20.000 observations, making it more manageable and relevant.

Finally, the dataset was split into a training set and a test set, with 70% of the data used for training and 30% for testing, following standard practice in data science to evaluate model performance.

By following these preprocessing steps, the dataset is thoroughly cleaned and prepared, ensuring that the analysis is based on reliable and well-structured data.

3 Data Analysis

R-squared indicates the proportion of the variance in the dependent variable that is predictable from the independent variables, providing a measure of how well our model fits the data.

The Root Mean Squared Error (RMSE) on the test set gives us an indication of the prediction accuracy of our model, with lower values indicating better performance.

In subsequent sections, these values will be compared with those obtained from other techniques to evaluate and identify the most effective predictive model.

By comparing these metrics across different models, we aim to identify which technique provides the best fit for our dataset and provides the most accurate predictions.

This comparative analysis will help us to understand the strengths and weaknesses of each approach and select the most appropriate model for our specific application.

3.1 Preliminary Analysis

For each variable it's been plotted the distribution with the purpose of seeing the behavior of each variable.

It's also been plotted a correlation matrix, it's so important for our analysis since it shows how the variables are correlates with each other.

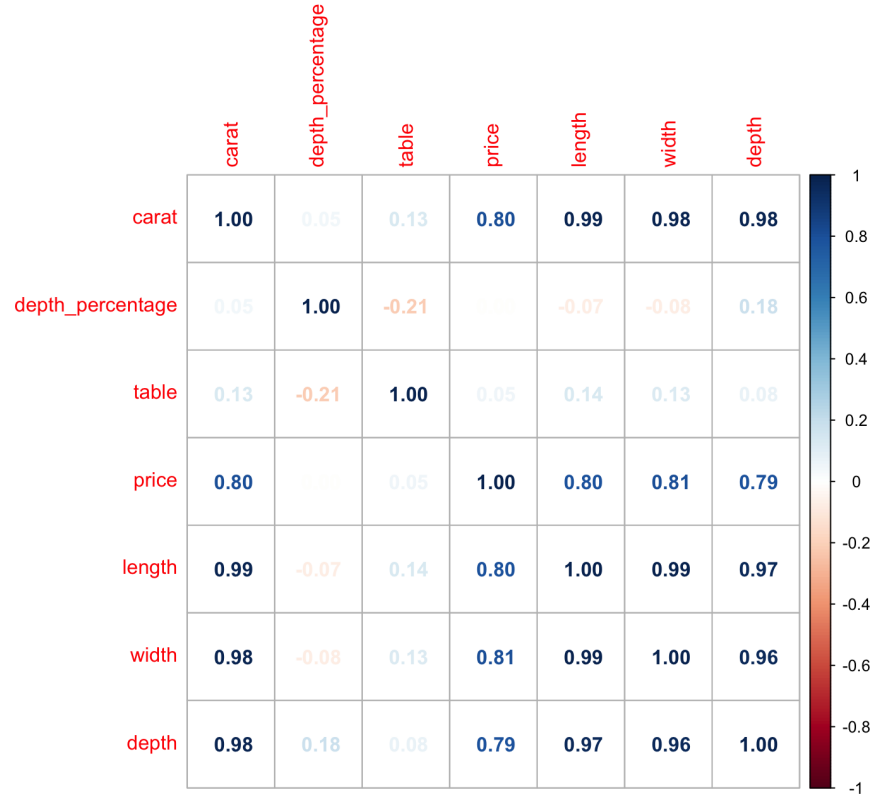


Figure 1: Correlation Matrix

4 Linear Regression

4.1 Introduction and Preprocessing Steps

Linear Regression is the first technique applied to the dataset. It is a fundamental method for modeling the relationship between variables.

In this project, it has been employed with the aim of understanding how various factors such as carat, cut, clarity, and color affect the output variable: the price of the diamond. By using Linear Regression, we can quantify the influence of each attribute on the diamond's price and identify which characteristics are the most significant predictors. Before applying this technique, the dataset was divided into two parts: one for training (70%) and another for testing (30%).

4.2 Linear Regression

After performing the linear regression on the dataset, it has been found that most of the regressors are relevant for the model. Among the regressors, the least relevant coefficients are depth_percentage and length.

R^2	RMSE
0.8878	0.5548

Table 3: R squared and test RMSE for Linear Regression

In the table above, we can see the obtained values of R-squared and the test RMSE using this technique.

4.3 Residuals Analysis

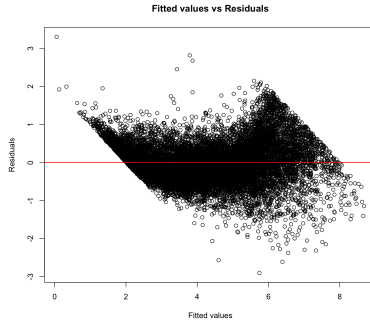


Figure 2: Fitted Values VS Residuals Values

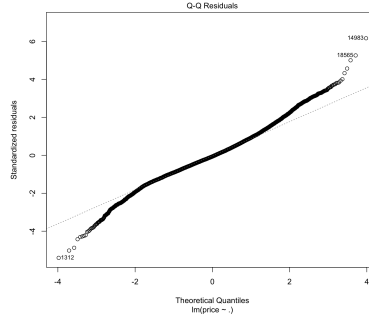


Figure 3: Q-Q Plot

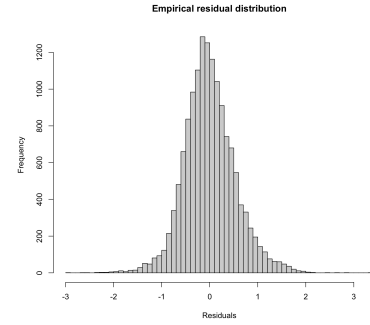


Figure 4: Frequency of Residual Values

Most of the residuals fall within the $[-1, 1]$ range, which translates to a range of -1000\$ to 1000\$. The model performs better on the central price values.

Despite the appearance in the third graph that the residuals might be normally distributed, they are actually not.

As shown in the second graph, the data do not perfectly follow the line; if they did, it would indicate a normal distribution.

The Shapiro test also confirms that the residuals are not normally distributed.

4.4 Interactions Term Between Physical Regressors

Upon observing how the physical dimensions (height, width, and depth) influenced the price of a diamond, a natural hypothesis emerged: while each of these three variables impacts the price in a certain way, the true determinant of a diamond's value is its overall shape, which is a combination of height, width, and depth.

We reasoned that considering these three variables together might perform better results.

Therefore, four new regressor were created, which are formed by the 2nd and 3rd order interactions between the three variables length, width and depth.

Subsequently, after adding these new coefficients, we applied linear regression to the updated model.

All new regressors are significant coefficients for the model.

The new model has led to improved results:

R^2	RMSE
0.8945	0.5328

Table 4: R squared and test RMSE for Linear Regression

Indeed, as illustrated, the second application of linear regression performs a lower RMSE value. Therefore, it can be concluded that the second model is better to the first.

To officially confirm this observation, an ANOVA test was performed. The test also confirmed that the second model performed better than the first one.

5 Best Subset Selection

The best subset selection technique was the next method applied to the model. For each model (from 1 to 27 regressors), it's been calculated the test RMSE and finally reported in the graph the best test RMSE for that number of regressor.

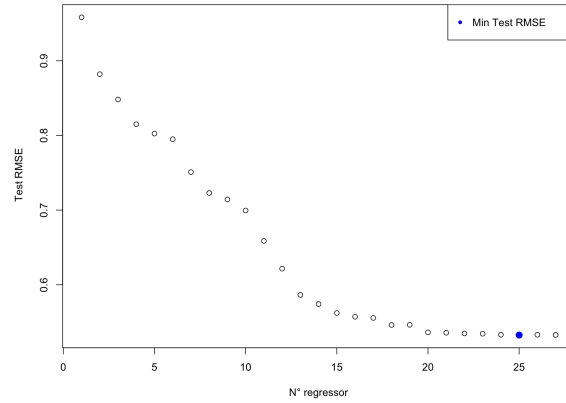


Figure 5: Best Subset Selection

By comparing the various RMSE values, it's been observed that the best model includes 25 regressors.

The regressors that were excluded are table and depth_percentage.

RMSE
0.5326

6 Ridge and Lasso Regressions

Ridge regression and lasso regression were applied to the dataset.

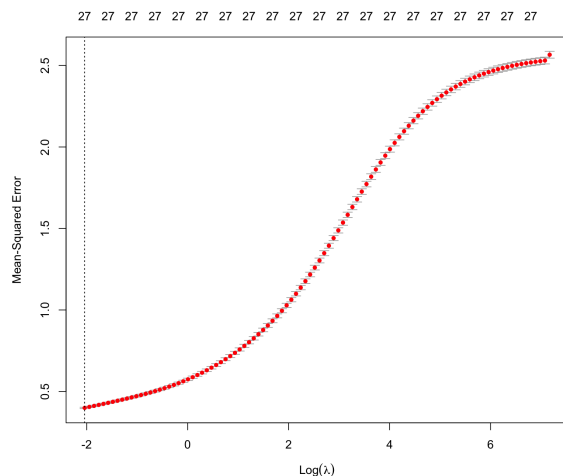


Figure 6: Ridge - Best Lambda

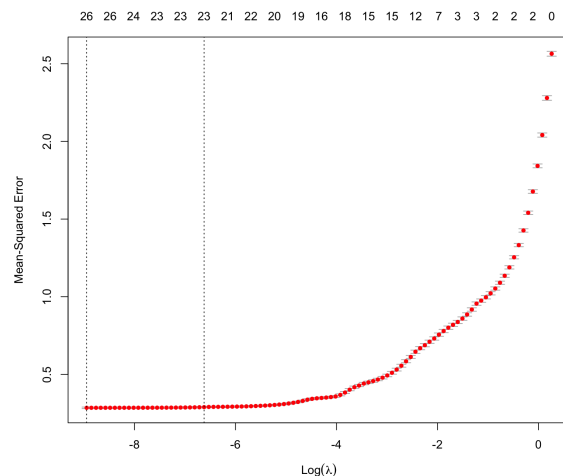


Figure 7: Lasso - Best Lambda

These techniques were not expected to perform well because both methods aim to reduce the variance of the model by shrinking insignificant coefficients to zero, or very close to zero.

The issue with our model, however, is that most of the coefficients are significant. Consequently, reducing the less significant coefficients to zero increases the bias (which is the goal of these techniques), but in this case, the reduction in variance does not compensate for the increase in bias.

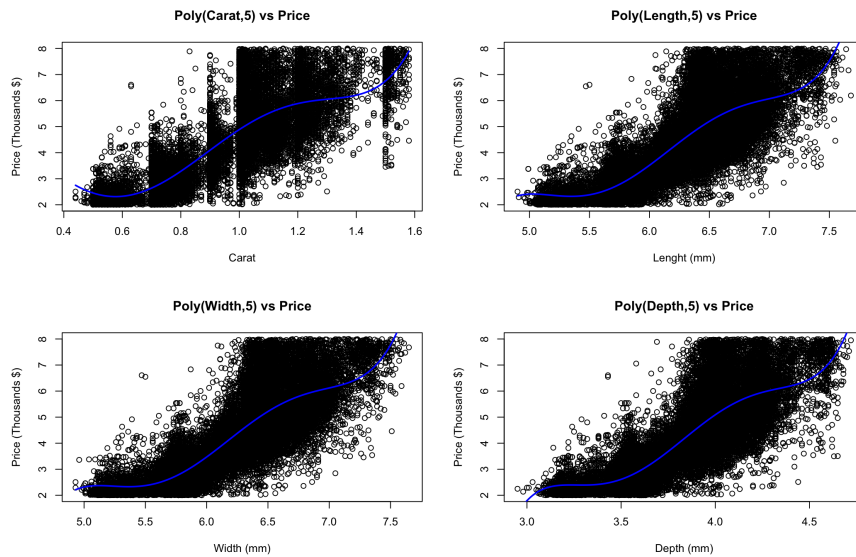
To confirm this, we also point out that the best lambda chosen by both models is always the smallest of the available values.

Ridge Regression		Lasso Regression	
RMSE	Best Lambda	RMSE	Best Lambda
0.6561	0.1297	0.5466	0.0001

Table 5: Comparison of RMSE values for Ridge and Lasso Regression

7 Polynomial Regression

Polynomial regression was applied to the model using different orders for various variables.



Specifically, a polynomial of order 5 was chosen for Carat, Length, and Width. For Table and for Depth_Percentage, a polynomial of order 2 was selected. The resulting model achieved a test RMSE of 0.5216, which is a significant improvement compared to linear models.

8 GAM - Generalized Additive Models

In order to capture the non-linear relationship between the regressors and the response variable, a GAM model was implemented.

In particular, it was chosen to use smoothing splines with four degrees of freedom for carat, width, length, and depth; a smoothing splines with two degrees of freedom for table and a smoothing spline with five degrees of freedom for depth percentage.

The obtained test RMSE is 0.5151, indicating a notable improvement compared to previous models.

9 Regression Trees

A regression tree was applied to the model.

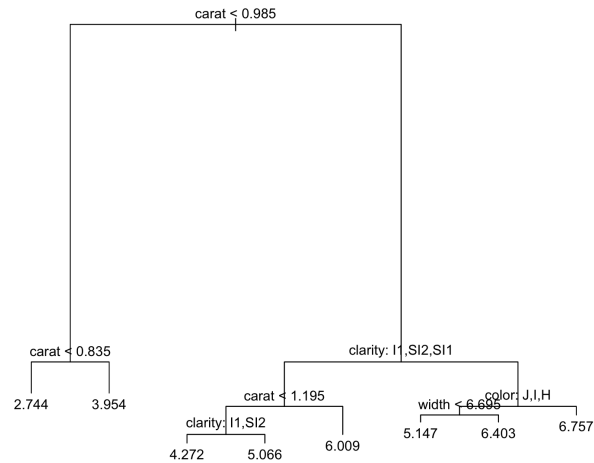


Figure 8: Regression Tree

Pruning was attempted using cross-validation pruning; however, no changes were made since the best number of leaves had already been achieved. The resulting model achieved a test RMSE of 0.7177.

10 Bagging and Random Forest

It's been performed a bagging method with 100 trees, the obtain test RMSE is 0.4015, is significantly better compared to the other models.

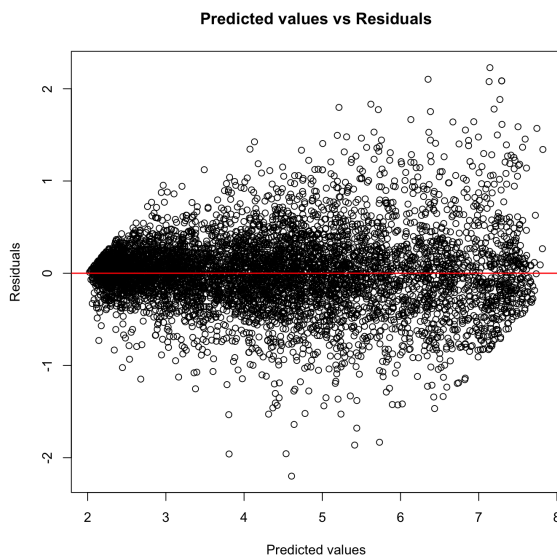


Figure 9: Bagging

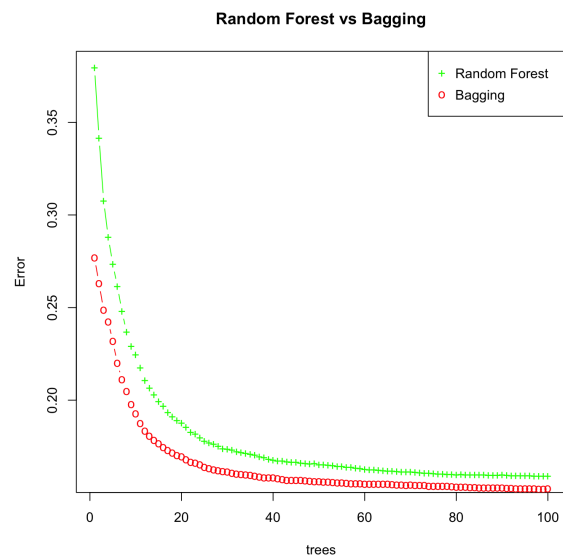


Figure 10: Comparison between Bagging and Random Forest

From the first graph it's possible to notice that the predictions compared with the linear models are more accurate.

In the second graph it's shown the comparison with random forest:

- The test RMSE of the bagging method is 0.4015
- The test RMSE of random forest is 0.4064

The RMSE are quite similar, both of them represent a good result but it's a little bit better the bagging ones.

11 Boosting

The last technique that it's been applied is boosting. Boosting is been performed with 5000 trees with $\lambda = 0.02$

With this method we had the best test RMSE obtained in all the project.

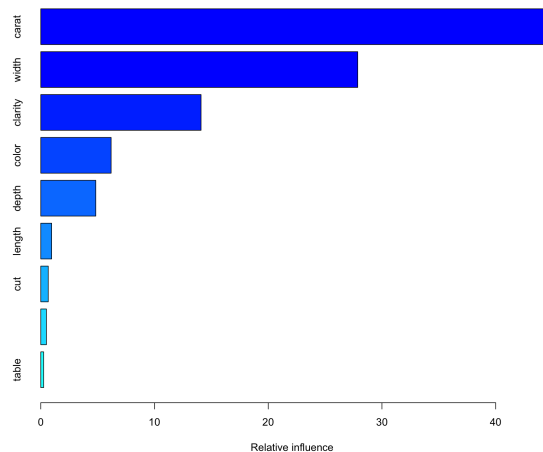


Figure 11: Variables Importance

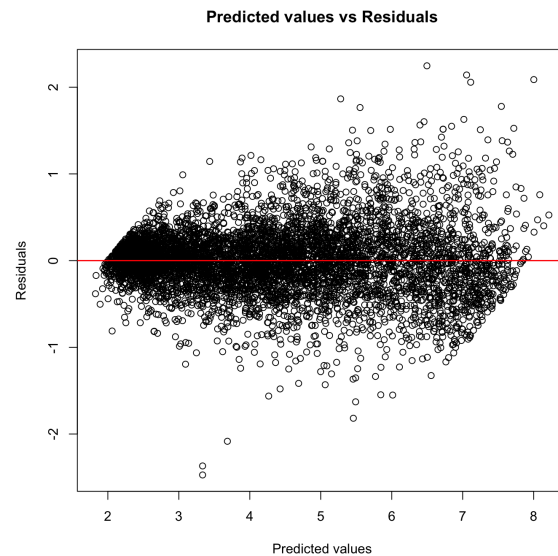


Figure 12: Boosting

The first graph shows the importance of the variables. The second graph shows that with boosting was again possible to notice how more accurate are the result comparing them with the previous models. The obtained test RMSE for this model is 0.3850, since now it's the best obtained RMSE.

12 Conclusions and Results Discussion

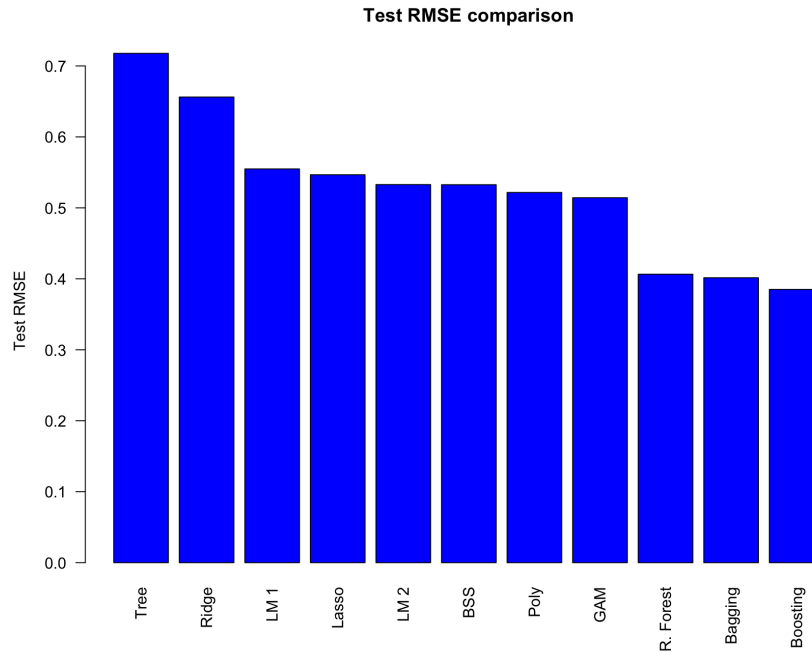


Figure 13: Comparison between all the applied models

The graph shows how different models performed in predicting diamond prices. Boosting stood out as the best, with an RMSE of \$385.

This indicates it was the most accurate in its predictions.

Interestingly, nonlinear models, which consider more complex relationships between factors, outperformed linear ones.

This suggests that diamond pricing isn't just about simple relationships between factors; there are more intricate dynamics at play.

The most important factors for predicting diamond prices were found to be Carat, Clarity, Width, and Color.

In summary, the Boosting model performed the best, nonlinear models were more accurate than linear ones, and Carat, Clarity, Width, and Color were identified as the most important factors for predicting diamond prices.

13 Classification

Since the conclusions of the first part are that Cut, Color, and Clarity are important factors, it has been decided to create a new output variable based on these three variables.

The new variable is called "Quality", it is a binary variable, so it can assume two possible values:

- High: Diamonds classified as "high" are those that have a high value of Color, Cut, and Clarity.
- Low: Diamonds classified as "low" are those that have a low value of Color, Cut, and

Clarity.

The objective of this analysis is to predict whether each diamond falls into the 'high' or 'low' category using the remaining regressors (all the ones we had but without color, clarity and cut).

The regressors considered are carat, price, depth percentage, table, width, length, and depth.

To balance the number of observations in each class, an undersampling technique was used, so the final dataset has 6037 entry. The scientific questions addressed in this study are:

* Which predictors have the greatest impact on classification?

* What is the nature of the boundary (linear, radial, or polynomial) that separates the two classes?

14 Logistic Regression

The first applied technique was logistic regression, which resulted in an accuracy of 89.06%.

It's been observed that the only significant coefficients are carat and price, especially watching the sign of coefficients it has been seen that increasing in price has a positive effect on the High classification, while increasing in carat has a negative effect.

Confusion Matrix:

	Low	High
Low	786	79
High	119	827

Table 6: Confusion Matrix for Logistic Regression

15 K-Nearest-Neighbour

The subsequent method applied was the k-nearest-neighbor (KNN) algorithm.

To find the best k parameter a 10-fold cross-validation approach was used. The best value for k was found to be 5.

However, KNN performed worse than logistic regression, yielding an accuracy of 86.52%, which is lower compared to that achieved with logistic regression. Confusion Matrix:

	Low	High
Low	768	107
High	137	799

Table 7: Confusion Matrix for Logistic Regression

16 Support Vector Machine

The last applied technique in this analysis was Support Vector Machines (SVM)

First of all it's been applied **SVM with a linear kernel**. It was necessary to deter-

mine the optimal cost parameter, resulting in the best cost being identified as 5. With this configuration, it's been achieved an accuracy of 89.50%. Subsequently, it's been explored **SVM with a radial kernel**. Employing cross-validation to fine-tune both the cost and gamma parameters, we found the optimal values to be cost = 5 and gamma = 0.5, yielding an accuracy of 90%, very similar to linear kernel SVM. Next, we investigated **SVM with a polynomial kernel**. Despite utilizing a degree of 3 and cost = 5, with gamma set to 0.5, this model performed worse than the previous ones, achieving an accuracy of 87.96%.

17 Final Considerations

Thanks to Logistic Regression it's clear that Carat and Price are the most significant variables, but can a SVM model (just with these two predictors) classify diamond's quality? Is linear kernel precise enough?

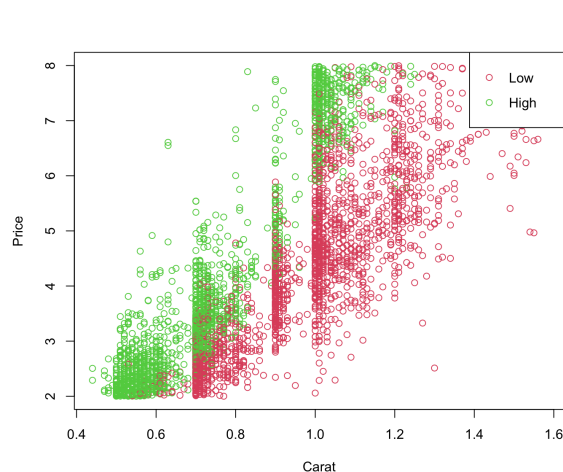


Figure 14: Carat vs Price with classes

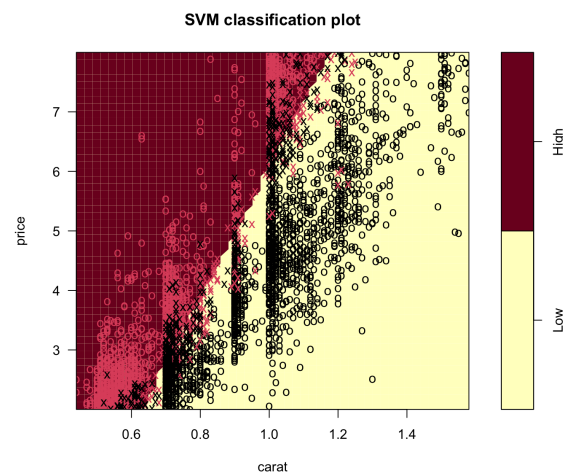


Figure 15: SVM Linear

An SVM linear kernel model with just these two predictors can effectively classify the quality of diamonds since the achieved accuracy is 89.61%, even better compared to previous model with all regressors. Furthermore we can see from the graphs that linear kernel is enough precise to correct classify Diamonds.

18 Conclusion

The crucial predictors necessary for accurately classifying the quality of each diamond are price and carat. SVM models with linear boundaries prove to be sufficiently accurate for this task.